1 Unconditional Non-linear denoiser

Random-feature model imposes a non-linear map on the input of the denoiser $\boldsymbol{y}$, however the denoiser output is still linear in this new non-linear feature map; let ψ be a non-linear function, i.e. sigmoid, softmax, etc, Θ be a $\mathbb{R}^{d\times k}$ random matrix, e.g Gaussian entries, ϵ be a $\mathbb{R}^d$ random bias; all are fixed i.e. we're not learning these here, as opposed to the way a full neural network does. Then:

$$\phi(\boldsymbol{y}) = \omega(\Theta\boldsymbol{y} + \epsilon)$$

And the denoiser w.r.t this new non-linear feature map is still:

$$\boldsymbol{D}_\theta(\phi(\boldsymbol{y}); \sigma, \boldsymbol{U}) = \boldsymbol{W}_\sigma\phi(\boldsymbol{y}) + \boldsymbol{b}_\sigma$$

where $\boldsymbol{W}_\sigma$ and $\boldsymbol{b}_\sigma$ are learned. We have:

$$\mathcal{L}_\sigma = \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}\|\boldsymbol{W}_\sigma\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \boldsymbol{b}_\sigma - \boldsymbol{x_0}\|_2^2$$

Note that the denoiser is linear w.r.t the feature map so most of our derivations from the linear case still applies. Setting derivative w.r.t $\boldsymbol{b}_\sigma$ to $\boldsymbol{0}$:

$$\begin{aligned}
\nabla_{\boldsymbol{b}_\sigma}\mathcal{L}_\sigma &= \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}\left[2(\boldsymbol{W}_\sigma\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \boldsymbol{b}_\sigma - \boldsymbol{x_0})\right] \\
0 &= 2(\boldsymbol{W}_\sigma\mu_\phi + \boldsymbol{b}_\sigma - \mu_{\boldsymbol{p_0}})
\end{aligned}$$

where $\mu_\phi = \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})]$. So:

$$\boldsymbol{b}_\sigma^* = \mu_{\boldsymbol{p_0}} - \boldsymbol{W}_\sigma\mu_\phi$$

Similarly:

$$\begin{aligned}
&\nabla_{\boldsymbol{W}_\sigma}\mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}\|\boldsymbol{W}_\sigma\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \boldsymbol{b}_\sigma - \boldsymbol{x_0}\|_2^2 \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}\left[2(\boldsymbol{W}_\sigma\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \boldsymbol{b}_\sigma - \boldsymbol{x_0})\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})^\top\right] \\
&= 2(\boldsymbol{W}_\sigma(\boldsymbol{\Sigma}_\phi + \mu_\phi\mu_\phi^\top) + (\mu_{\boldsymbol{p_0}} - \boldsymbol{W}_\sigma\mu_\phi)\mu_\phi - \mathbb{E}[\boldsymbol{x_0}\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})^\top]) \\
0 &= 2(\boldsymbol{W}_\sigma(\boldsymbol{\Sigma}_\phi) + \mathrm{Cov}(\boldsymbol{x_0}, \phi))
\end{aligned}$$

where $\boldsymbol{\Sigma}_\phi = \mathrm{Cov}(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}))$. So $\boldsymbol{W}_\sigma^* = \mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1}$. Then:

$$\begin{aligned}
\mathcal{L}_\sigma &= \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}\|\boldsymbol{W}_\sigma(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) - \mu_\phi) - (\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})\|_2^2 \\
&= \mathrm{Tr}(\mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\boldsymbol{W}_\sigma(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) - \mu_\phi)(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) - \mu_\phi)^\top\boldsymbol{W}_\sigma^\top] \\
&- 2\mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\boldsymbol{W}_\sigma(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) - \mu_\phi)(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})^\top] + \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})^\top]) \\
&= \mathrm{Tr}(\boldsymbol{W}_\sigma\boldsymbol{\Sigma}_{\phi(\boldsymbol{x_0}+\sigma\boldsymbol{Z})-\mu_\phi}\boldsymbol{W}_\sigma^\top - 2\boldsymbol{W}_\sigma\mathrm{Cov}(\boldsymbol{x_0}, \phi) + \boldsymbol{\Sigma}_{\boldsymbol{x_0}-\mu_{\boldsymbol{p_0}}}) \\
&= \mathrm{Tr}(\boldsymbol{W}_\sigma\boldsymbol{\Sigma}_\phi\boldsymbol{W}_\sigma^\top - 2\boldsymbol{W}_\sigma\mathrm{Cov}(\boldsymbol{x_0}, \phi) + \boldsymbol{\Sigma}_{\boldsymbol{p_0}}) \\
&= \mathrm{Tr}(\mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1}\boldsymbol{\Sigma}_\phi(\mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1})^\top - 2\mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1}\mathrm{Cov}(\boldsymbol{x_0}, \phi) + \boldsymbol{\Sigma}_{\boldsymbol{p_0}}) \\
&= \mathrm{Tr}(\boldsymbol{\Sigma}_{\boldsymbol{p_0}} - \mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1}\mathrm{Cov}(\phi, \boldsymbol{x_0}))
\end{aligned}$$

where $\mathrm{Cov}(\boldsymbol{x_0}, \phi) = \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) - \mu_\phi)(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})^\top]$

Now if ϕ is Gaussian square-integrable w.r.t the Gaussian measure, we can use Hermite polynomials to approximate it. Recall:

$$\mathrm{He}_n = (-1)^n e^{x^2/2}\frac{d^n}{dx^n}(e^{-x^2/2})$$

The first few look like:

$$\begin{aligned}
\mathrm{He}_0 &= 1 \\
\mathrm{He}_1 &= x - 1 \\
\mathrm{He}_2 &= x^2 - 1 \\
\mathrm{He}_3 &= x^3 - 3x \\
\mathrm{He}_4 &= x^4 - 6x^2 + 3 \\
&\vdots
\end{aligned}$$

We can show that:

$$\mathrm{He}_{n+1}(x) = x\mathrm{He}_n(x) - n\mathrm{He}_{n-1}(x)$$

These polynomials are the orthogonal basis of $\mathcal{P}(x)$, the space of all polynomials in x; their orthogonality is under the inner-product of the space which is with respect to the Gaussian measure $\langle \cdot, \cdot \rangle_\Phi$, given as $\langle p, q\rangle_\varphi = \int p(x)q(x)\varphi(x)dx$ where $\varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$. In other words, $\langle p, q\rangle_\varphi = \mathbb{E}_{Z\sim\mathcal{N}(0,1)}[p(Z)q(Z)]$. Then:

$$f(x) = \sum_n \frac{1}{n!}\langle f(x), \mathrm{He}_n(x)\rangle_\varphi \mathrm{He}_n(x)$$

Denote $\frac{1}{n!}\langle f(x), \mathrm{He}_n(x)\rangle_\varphi = c_n(x)$
First consider μ_ϕ:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})] &= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\sum_n c_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})\mathrm{He}_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[\sum_n c_n \mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[\mathrm{He}_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[\sum_n \langle \mathrm{He}_0, c_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})\mathrm{He}_n\rangle_\varphi]
\end{aligned}$$

We want to expand the closed-form of $\mathcal{L}_\sigma$. Here if we Hermite-expand on the non-linear $\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z})$ then our Hermite polynomials would look like $\mathrm{He}_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})$ and coefficients like $c_n(\boldsymbol{x_0} + \sigma\boldsymbol{Z})$, which is not ideal for simplifying any further since we will be dealing with $\boldsymbol{x_0} + \sigma\boldsymbol{Z}$ which is not Gaussian unless $\boldsymbol{x_0}$ is Gaussian. So first condition on $\boldsymbol{x_0}$ first and then expand. Consider the case ϕ is a component-wise non-linear function; then we can get around $\phi(\boldsymbol{y}) = \omega(\Theta\boldsymbol{y} + \xi)$ by building it row by row. Let θ_j be the j^{th} row of Θ, and conditional on $\boldsymbol{x_0}$:

$$\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) = \omega(\theta_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \epsilon_j) = \omega(\theta_j^\top\boldsymbol{x_0} + \sigma\theta_j^\top\boldsymbol{Z} + \epsilon_j) \coloneqq \omega(M_j(\boldsymbol{x_0}) + s_j\xi);$$
$$\text{where} \qquad s_j = \sigma\|\theta_j\|, \quad \xi = \frac{\theta_j^\top\boldsymbol{Z}}{\|\theta_j\|} \sim \mathcal{N}(0,1), \quad M_j(\boldsymbol{x_0}) = \theta_j^\top\boldsymbol{x_0} + \epsilon_j$$

Conditional on $\boldsymbol{x_0}$, $M_j(\boldsymbol{x_0})$ is just a constant; and s_j is also a constant. Then ω really is a function of ξ; let $f(\xi) : \xi \mapsto \omega(M_j(\boldsymbol{x_0}) + s_j\xi)$. Now we Hermite-expand $f(\xi)$:

$$\omega(M_j(\boldsymbol{x_0}) + s_j\xi) = \sum_{n=0} c_n(M_j(\boldsymbol{x_0}), s_j)\mathrm{He}_n(\xi), \qquad c_n(M_j(\boldsymbol{x_0}), s_j) = \frac{1}{n!}\langle f(\xi), \mathrm{He}_n(\xi)\rangle_\varphi = \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[f(\xi)\mathrm{He}_n(\xi)]$$

Note that $c_n(M_j(\boldsymbol{x_0}), s_j)$ is a constant w.r.t ξ since ξ would be integrated out. Also that $\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\mathrm{He}_i(\xi)\mathrm{He}_j(\xi)] = 0$ by orthogonality. Then by Tower property:

$$\begin{aligned}
\mathbb{E}[\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z})] &= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0}}[\mathbb{E}[\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0}}[\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\sum_{n=0} c_n(M_j(\boldsymbol{x_0}), s_j)\mathrm{He}_n(\xi)]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0}}[\sum_{n=0} c_n(M_j(\boldsymbol{x_0}), s_j)\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\mathrm{He}_n(\xi)\mathrm{He}_0(\xi)]] \qquad (\text{since } \mathrm{He}_0 = 1) \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0}}[c_0(M_j(\boldsymbol{x_0}), s_j)]
\end{aligned}$$

Rewrite $c_0(M_j(\boldsymbol{x_0}), s_j) \coloneqq g_j(\boldsymbol{x_0})$ since g_j varies with $\boldsymbol{x_0}$:

$$\begin{aligned}
g_j(\boldsymbol{x_0}) &= \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[f(\xi)\mathrm{He}_0(\xi)] \\
&= \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\omega(M_j(\boldsymbol{x_0}) + s_j\xi)] \\
&= \int \omega(M_j(\boldsymbol{x_0}) + s_j\xi)\frac{e^{-\xi^2/2}}{\sqrt{2\pi}}d\xi
\end{aligned}$$

Let $u = s_j\xi$, so $\xi = \frac{u}{s_j}$ and $d\xi = \frac{du}{s_j}$:

$$g_j(\boldsymbol{x_0}) = \int \omega(M_j(\boldsymbol{x_0}) + u)\frac{e^{-u^2/2s_j^2}}{s_j\sqrt{2\pi}}du$$

This integral is the convolution of function ω and the Gaussian $\mathcal{N}(0, s_j^2)$, evaluated at the local point $M_j(\boldsymbol{x_0})$, i.e.

$$g_j(\boldsymbol{x_0}) = (\omega * \mathcal{N}_{s_j^2})(M_j(\boldsymbol{x_0}))$$

The interpretation is that this is a function ω blurred by a Gaussian kernel of width s_j, then evaluated at the point $M_j(\boldsymbol{x_0})$. Now consider $\mathrm{Cov}(\boldsymbol{x_0}, \phi)_{ij}$ at row i^{th} and column j^{th}:

$$\begin{aligned}
\mathrm{Cov}(\boldsymbol{x_0}, \phi)_{ij} &= \mathbb{E}_{\boldsymbol{x_0}\sim p_0, \boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})(\phi(\boldsymbol{x_0} + \sigma \boldsymbol{Z}))_j] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})(\phi(\boldsymbol{x_0} + \sigma \boldsymbol{Z}))_j \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\phi(\boldsymbol{x_0} + \sigma \boldsymbol{Z}))_j \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j(\boldsymbol{x_0})]
\end{aligned}$$

Note that $g_j(\boldsymbol{x_0}) = c_0(M_j(\boldsymbol{x_0}), s_j)$ where $M_j(\boldsymbol{x_0}) = \Theta_j^\top \boldsymbol{x_0} + \epsilon_j$ so really $g_j(\boldsymbol{x_0})$ only depends on $u_j \coloneqq \Theta_j^\top \boldsymbol{x_0} + \epsilon_j$, so rewrite $g_j(\boldsymbol{x_0}) = \psi(\Theta_j^\top \boldsymbol{x_0} + \epsilon_j) = \psi(u_j)$ where $\psi(u) = c_0(u, s_j)$. Now by Tower property:

$$\begin{aligned}
\mathrm{Cov}(\boldsymbol{x_0}, \phi)_{ij} &= \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j(\boldsymbol{x_0})] \\
&= \mathbb{E}_{u_j}[\mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j(\boldsymbol{x_0}) \mid u_j]] \\
&= \mathbb{E}_{u_j}[\mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j(\boldsymbol{x_0}) \mid u_j]] \\
&= \mathbb{E}_{u_j}[g_j(u_j)\mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i}) \mid u_j]]
\end{aligned}$$

Define $r_i(u_j) = \mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i}) \mid u_j]]$, which is really a regression coefficient of $\boldsymbol{x_0}$ onto its own linear projection. All this do is turning a multi-dimensional integral $\mathbb{E}_{\boldsymbol{x_0}\sim p_0}$ into an 1-D integral $\mathbb{E}_{u_j}$.

Now consider $\boldsymbol{\Sigma}_\phi$; recall that ω is component-wise so we're allowed to break ϕ down into its entries.

$$\boldsymbol{\Sigma}_{\phi,ij} = \mathrm{Cov}(\phi_i, \phi_j) = \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i\phi_j] - \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i]\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_j]$$

Consider the first expectation term; by the Tower property:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i\phi_j] &= \mathbb{E}_{\boldsymbol{x_0}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_i(\boldsymbol{x_0} + \sigma\boldsymbol{Z})\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}}\big[\mathrm{Cov}(\phi_i, \phi_j \mid \boldsymbol{x_0}) + \mathbb{E}_{\boldsymbol{Z}}[\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) \mid \boldsymbol{x_0}]\mathbb{E}_{\boldsymbol{Z}}[\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) \mid \boldsymbol{x_0}]\big] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}(\phi_i, \phi_j \mid \boldsymbol{x_0})] + \mathbb{E}_{\boldsymbol{x_0}}[g_i(\boldsymbol{x_0})g_j(\boldsymbol{x_0})]
\end{aligned}$$

Now:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i]\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_j] &= \mathbb{E}_{\boldsymbol{x_0}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_i \mid \boldsymbol{x_0}]]\mathbb{E}_{\boldsymbol{x_0}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_i \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[g_i(\boldsymbol{x_0})]\mathbb{E}_{\boldsymbol{x_0}}[g_j(\boldsymbol{x_0})]
\end{aligned}$$

So:

$$\begin{aligned}
\boldsymbol{\Sigma}_{\phi,ij} &= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i\phi_j] - \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_i]\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z}}[\phi_j] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}_{\boldsymbol{Z}}(\phi_i, \phi_j \mid \boldsymbol{x_0})] + \mathbb{E}_{\boldsymbol{x_0}}[g_i(\boldsymbol{x_0})g_j(\boldsymbol{x_0})] - \mathbb{E}_{\boldsymbol{x_0}}[g_i(\boldsymbol{x_0})]\mathbb{E}_{\boldsymbol{x_0}}[g_j(\boldsymbol{x_0})] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}_{\boldsymbol{Z}}(\phi_i, \phi_j \mid \boldsymbol{x_0})] + \mathrm{Cov}_{\boldsymbol{x_0}}(g_i(\boldsymbol{x_0}), g_j(\boldsymbol{x_0}))
\end{aligned}$$

Consider the term $\mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}(\phi_i, \phi_j \mid \boldsymbol{x_0})]$:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}_{\boldsymbol{Z}}(\phi_i, \phi_j \mid \boldsymbol{x_0})] &= \mathbb{E}_{\boldsymbol{x_0}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_i\phi_j \mid \boldsymbol{x_0}] - \mathbb{E}_{\boldsymbol{Z}}[\phi_i \mid \boldsymbol{x_0}]\mathbb{E}_{\boldsymbol{Z}}[\phi_j \mid \boldsymbol{x_0}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_i\phi_j \mid \boldsymbol{x_0}] - c_0(M_i(\boldsymbol{x_0}), s_i)c_0(M_j(\boldsymbol{x_0}), s_j)]
\end{aligned}$$

Recall Hermite-expansion of ϕ conditional on $\boldsymbol{x_0}$:

$$\phi_i = \sum_{n=0} c_n(M_i(\boldsymbol{x_0}), s_i)\mathrm{He}_n(\xi_i), \quad \phi_j = \sum_{m=0} c_m(M_j(\boldsymbol{x_0}), s_j)\mathrm{He}_m(\xi_j);$$

So:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}[\phi_i\phi_j \mid \boldsymbol{x_0}] &= \mathbb{E}_{\boldsymbol{Z}}\big[\sum_{n=0} c_n(M_i(\boldsymbol{x_0}), s_i)\mathrm{He}_n(\xi_i) \sum_{m=0} c_m(M_j(\boldsymbol{x_0}), s_j)\mathrm{He}_m(\xi_j)\big] \\
&= \sum_{n,m} c_n(M_i(\boldsymbol{x_0}), s_i)c_m(M_j(\boldsymbol{x_0}), s_j)\mathbb{E}\big[\mathrm{He}_n(\xi_i)\mathrm{He}_m(\xi_j)\big]
\end{aligned}$$

Using the Mehler product identity for correlated Gaussian:

$$\mathbb{E}\big[\mathrm{He}_n(\xi_i)\mathrm{He}_m(\xi_j)\big] = n!\rho_{ij}^n \delta_{nm}$$

where $\rho_{ij} = \mathrm{Corr}(\xi_i, \xi_j) = \frac{\theta_i^\top \theta_j}{\|\theta_i\|\|\theta_j\|}$ and δ_{nm} is the Kronecker delta. So:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}[\phi_i\phi_j \mid \boldsymbol{x_0}] &= \sum_n \sum_m n! c_n(M_i(\boldsymbol{x_0}), s_i) c_m(M_j(\boldsymbol{x_0}), s_j) \rho_{ij}^n \delta_{nm} \\
&= \sum_n n!\rho_{ij}^n c_n(M_i(\boldsymbol{x_0}), s_i) \sum_m c_m(M_j(\boldsymbol{x_0}), s_j)\delta_{nm} \\
&= \sum_n n!\rho_{ij}^n c_n(M_i(\boldsymbol{x_0}), s_i) c_n(M_j(\boldsymbol{x_0}), s_j)
\end{aligned}$$

Therefore:

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x_0}}[\mathrm{Cov}_{\boldsymbol{Z}}(\phi_i, \phi_j \mid \boldsymbol{x_0})] &= \mathbb{E}_{\boldsymbol{x_0}}[\sum_n n!\rho_{ij}^n c_n(M_i(\boldsymbol{x_0}), s_i) c_n(M_j(\boldsymbol{x_0}), s_j) - c_0(M_i(\boldsymbol{x_0}), s_i) c_0(M_j(\boldsymbol{x_0}), s_j)] \\
&= \mathbb{E}_{\boldsymbol{x_0}}[\sum_{n\geq 1} n!\rho_{ij}^n c_n(M_i(\boldsymbol{x_0}), s_i) c_n(M_j(\boldsymbol{x_0}), s_j)]
\end{aligned}$$

So:

$$\boldsymbol{\Sigma}_{\phi,ij} = \mathrm{Cov}_{\boldsymbol{x_0}}(g_i(\boldsymbol{x_0}), g_j(\boldsymbol{x_0})) + \mathbb{E}_{\boldsymbol{x_0}}[\sum_{n\geq 1} n!\rho_{ij}^n c_n(M_i(\boldsymbol{x_0}), s_i) c_n(M_j(\boldsymbol{x_0}), s_j)]$$

To summarize:

$$\mathcal{L}_\sigma = \mathrm{Tr}(\boldsymbol{\Sigma}_{\boldsymbol{p_0}} - \mathrm{Cov}(\boldsymbol{x_0}, \phi)\boldsymbol{\Sigma}_\phi^{-1}\mathrm{Cov}(\phi, \boldsymbol{x_0}))$$

where the entries of $\boldsymbol{\Sigma}_\phi$ is defined as above and the entries of $\mathrm{Cov}(\boldsymbol{x_0}, \phi)$ is

$$\mathrm{Cov}(\boldsymbol{x_0}, \phi)_{ij} = \mathbb{E}_{u_j}[g_j(u_j)\mathbb{E}_{\boldsymbol{x_0}\sim p_0}[(\boldsymbol{x_{0}}_{,i} - \mu_{\boldsymbol{p_0},i}) \mid u_j]]$$

2 Conditional Non-linear denoiser

In the conditional case, if we treat $(\boldsymbol{y}, \boldsymbol{U})$ as a single joint input vector where $\phi^{\boldsymbol{U}}(\boldsymbol{y}, \boldsymbol{U}) = \omega(\Theta\boldsymbol{y} + \Gamma\boldsymbol{U} + \epsilon)$ is a random feature map then the loss has the same form as before except that:

$$\begin{aligned}
\mu_\phi^{\boldsymbol{U}} &= \mathbb{E}[\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U})] \\
\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}} &= \mathrm{Cov}(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U})) \\
\mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}}) &= \mathbb{E}[(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})(\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}) - \mu_\phi^{\boldsymbol{U}})^\top]
\end{aligned}$$

and so:

$$\mathcal{L}_{\sigma,\boldsymbol{U}} = \mathrm{Tr}(\boldsymbol{\Sigma}_{\boldsymbol{p_0}} - \mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}})(\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}})^{-1}\mathrm{Cov}(\phi^{\boldsymbol{U}}, \boldsymbol{x_0}))$$

We want to expand this more. Carrying over from the unconditional case, let θ_j be the j^{th} row of Θ, γ_j be the j^{th} row of Γ, and conditional on $\boldsymbol{x_0}$:

$$\begin{aligned}
&\phi_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) = \omega(\theta_j(\boldsymbol{x_0} + \sigma\boldsymbol{Z}) + \gamma_j\boldsymbol{U} + \epsilon_j) = \omega(\theta_j^\top\boldsymbol{x_0} + \sigma\theta_j^\top\boldsymbol{Z} + \gamma_j^\top\boldsymbol{U} + \epsilon_j) \coloneqq \omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + s_j\xi); \\
&\text{where} \qquad s_j = \sigma\|\theta_j\|, \quad \xi = \frac{\theta_j^\top\boldsymbol{Z}}{\|\theta_j\|} \sim \mathcal{N}(0,1), \quad M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = \theta_j^\top\boldsymbol{x_0} + \gamma_j\boldsymbol{U} + \epsilon_j
\end{aligned}$$

We're given $\boldsymbol{U}$ and conditional on $\boldsymbol{x_0}$, ω really is a function of ξ; let $f(\xi) : \xi \mapsto \omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + s_j\xi)$. Hermite-expand $f(\xi)$:

$$\omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + s_j\xi) = \sum_{n=0} c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)\mathrm{He}_n(\xi), \qquad c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j) = \frac{1}{n!}\langle f(\xi), \mathrm{He}_n(\xi)\rangle_\varphi = \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[f(\xi)\mathrm{He}_n(\xi)]$$

Note that $c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)$ is a constant w.r.t ξ since ξ would be integrated out. Also that $\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\mathrm{He}_i(\xi)\mathrm{He}_j(\xi)] = 0$ by orthogonality. Then by Tower property:

$$
\begin{aligned}
\mathbb{E}[\phi_j^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U})] &= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0},\boldsymbol{U}}[\mathbb{E}_{\boldsymbol{Z}}[\phi_j^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}) \mid \boldsymbol{x_0}, \boldsymbol{U}]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0},\boldsymbol{U}}[\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\sum_{n=0} c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)\mathrm{He}_n(\xi)]] \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0},\boldsymbol{U}}[\sum_{n=0} c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)\mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\mathrm{He}_n(\xi)\mathrm{He}_0(\xi)]] \qquad (\text{since } \mathrm{He}_0 = 1) \\
&= \mathbb{E}_{\boldsymbol{x_0}\sim\boldsymbol{p_0},\boldsymbol{U}}[c_0(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)]
\end{aligned}
$$

Rewrite $c_0(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j) \coloneqq g_j^{\boldsymbol{U}}(\boldsymbol{x_0})$ since $g_j^{\boldsymbol{U}}$ varies with $\boldsymbol{x_0}$ and $\boldsymbol{U}$ was given:

$$
\begin{aligned}
g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) &= \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[f(\xi)\mathrm{He}_0(\xi) \mid \boldsymbol{x_0}, \boldsymbol{U}] \\
&= \mathbb{E}_{\xi\sim\mathcal{N}(0,1)}[\omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + s_j\xi)] \\
&= \int \omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + s_j\xi)\frac{e^{-\xi^2/2}}{\sqrt{2\pi}}d\xi
\end{aligned}
$$

Let $u = s_j\xi$, so $\xi = \frac{u}{s_j}$ and $d\xi = \frac{du}{s_j}$:

$$
g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = \int \omega(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) + u)\frac{e^{-u^2/2s_j^2}}{s_j\sqrt{2\pi}}du
$$

This integral is the convolution of function ω and the Gaussian $\mathcal{N}(0, s_j^2)$, evaluated at the local point $M_j^{\boldsymbol{U}}(\boldsymbol{x_0})$, i.e.

$$
g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = (\omega * \mathcal{N}_{s_j^2})(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}))
$$

And so it follows that:

$$
\boldsymbol{\Sigma}_{\phi,ij}^{\boldsymbol{U}} = \mathrm{Cov}_{\boldsymbol{x_0},\boldsymbol{U}}(g_i^{\boldsymbol{U}}(\boldsymbol{x_0}), g_j^{\boldsymbol{U}}(\boldsymbol{x_0})) + \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[\sum_{n\geq 1} n!\rho_{ij}^n c_n(M_i^{\boldsymbol{U}}(\boldsymbol{x_0}), s_i)c_n(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)] \tag{2.1}
$$

Now consider $\mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}})_{ij}$ at row i^{th} and column j^{th}:

$$
\begin{aligned}
\mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}})_{ij} &= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})(\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}))_j] \\
&= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})(\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}))_j \mid \boldsymbol{x_0}, \boldsymbol{U}]] \\
&= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})\mathbb{E}_{\boldsymbol{Z}\sim\mathcal{N}(0,\boldsymbol{I})}[(\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}))_j \mid \boldsymbol{x_0}, \boldsymbol{U}]] \\
&= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j^{\boldsymbol{U}}(\boldsymbol{x_0})]
\end{aligned}
$$

Note that $g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = c_0(M_j^{\boldsymbol{U}}(\boldsymbol{x_0}), s_j)$ where $M_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = \theta_j^\top\boldsymbol{x_0} + \gamma_j^\top\boldsymbol{U} + \epsilon_j$ so really $g_j^{\boldsymbol{U}}(\boldsymbol{x_0})$ only depends on $h_j \coloneqq \theta_j^\top\boldsymbol{x_0} + \gamma_j^\top\boldsymbol{U} + \epsilon_j$, so rewrite $g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) = \psi(\theta_j^\top\boldsymbol{x_0} + \gamma_j^\top\boldsymbol{U} + \epsilon_j) = \psi(h_j)$ where $\psi(h) = c_0(h, s_j)$. Now by Tower property:

$$
\begin{aligned}
\mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}})_{ij} &= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j^{\boldsymbol{U}}(\boldsymbol{x_0})] \\
&= \mathbb{E}_{h_j}[\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i})g_j^{\boldsymbol{U}}(\boldsymbol{x_0}) \mid h_j]] \\
&= \mathbb{E}_{h_j}[g_j^{\boldsymbol{U}}(h_j)\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i}) \mid h_j]]
\end{aligned}
$$

Define $r_i(h_j) = \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}[(\boldsymbol{x}_{\boldsymbol{0},i} - \mu_{\boldsymbol{p_0},i}) \mid h_j]$, which is really a regression coefficient of $[\boldsymbol{x_0}, \boldsymbol{U}]$ onto its own linear projection $h_j = [\theta_j, \gamma_j]^\top[\boldsymbol{x_0}, \boldsymbol{U}]$ plus noise. All this do is turning a multi-dimensional integral $\mathbb{E}_{\boldsymbol{x_0},\boldsymbol{U}}$ into an 1-D integral $\mathbb{E}_{h_j}$.

$\mathcal{L}_{\sigma,\boldsymbol{U}}$ capture information of how $\boldsymbol{x_0}$ relates to $\boldsymbol{U}$ in a non-linear way.

Note: Recall that what follows is only valid when ω is component-wise, so the formulas we derived would not apply to ω being a function like softmax. In such case, $\mu_\phi^{\boldsymbol{U}} = \mathbb{E}[\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U})]$, $\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}} = \mathrm{Cov}(\phi(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}))$ and $\mathrm{Cov}(\boldsymbol{x_0}, \phi^{\boldsymbol{U}}) = \mathbb{E}[(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})(\phi^{\boldsymbol{U}}(\boldsymbol{x_0} + \sigma\boldsymbol{Z}, \boldsymbol{U}) - \mu_\phi^{\boldsymbol{U}})^\top]$ would have to be left in these forms and can't be broken down any further, and remain integrals over all of $\boldsymbol{x_0} \sim \boldsymbol{p_0}, \boldsymbol{Z}, \boldsymbol{U}$. Though in the case of softmax, if somehow the denominator can be asymptotically approximated as some quantity that is independent of each of the entries that sum to it, e.g. by LLN, then ω can still be considered component-wise and our derivations apply.

$$\begin{aligned}
I(X;U) &= \frac{1}{2}\int_0^\infty \frac{dt}{t^2}(\text{MMSE }[X \mid Y_t] - \text{MMSE }[X \mid Y_t, U]) \\
&\approx \frac{1}{2}\int_0^\infty \frac{d\sigma}{\sigma^3}(\mathcal{L}_\sigma - \mathcal{L}_{\sigma,\boldsymbol{U}}) \\
&\approx \frac{1}{2}\int_0^\infty \frac{d\sigma}{\sigma^3}\text{Tr}(\text{Cov}(\boldsymbol{x_0},\phi^{\boldsymbol{U}})(\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}})^{-1}\text{Cov}(\phi^{\boldsymbol{U}},\boldsymbol{x_0}) - \text{Cov}(\boldsymbol{x_0},\phi)\boldsymbol{\Sigma}_\phi^{-1}\text{Cov}(\phi,\boldsymbol{x_0}))
\end{aligned}$$

Note: $\text{Cov}(\boldsymbol{x_0},\phi)\boldsymbol{\Sigma}_\phi^{-1}\text{Cov}(\phi,\boldsymbol{x_0})$ is the explained variance of the best possible linear regression of $\boldsymbol{x_0}$ onto ϕ.

so $\text{Cov}(\boldsymbol{x_0},\phi^{\boldsymbol{U}})(\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}})^{-1}\text{Cov}(\phi^{\boldsymbol{U}},\boldsymbol{x_0}) \geq \text{Cov}(\boldsymbol{x_0},\phi)\boldsymbol{\Sigma}_\phi^{-1}\text{Cov}(\phi,\boldsymbol{x_0})$ if $\text{span}(\phi) \subseteq \text{span}(\phi^{\boldsymbol{U}})$. Otherwise, the integrand is not nonnegative.

3 Jointly Gaussian $(\boldsymbol{x_0}, \boldsymbol{U})$

In the case of $(\boldsymbol{x_0}, \boldsymbol{U})$ jointly Gaussian, we can simplify the expression even more as this buys us Stein's lemma. Now $l_j := \theta_j^\top \boldsymbol{x_0} + \gamma_j^\top \boldsymbol{U} + \epsilon_j + s_j\xi$ is Gaussian, and equivalently $l_j = [\theta_j, \gamma_j]^\top[\boldsymbol{x_0}, \boldsymbol{U}] + \epsilon_j + s_j\xi$. Let the co-variance between $\boldsymbol{x_0}$ and $\boldsymbol{U}$ be $C_{\boldsymbol{xU}}$. Then l_j has mean and variance:

$$\begin{aligned}
\tilde{m_j} &= \theta_j\mu_{\boldsymbol{x_0}} + \gamma_j\mu_{\boldsymbol{U}} \\
\tilde{S_j^2} &= \text{Var}([\theta_j,\gamma_j]^\top[\boldsymbol{x_0},\boldsymbol{U}]) + \text{Var}(s_j\xi) \\
&= [\theta_j,\gamma_j]^\top \begin{bmatrix} \boldsymbol{\Sigma_{p_0}} & C_{\boldsymbol{xU}} \\ C_{\boldsymbol{Ux}} & \boldsymbol{\Sigma_U} \end{bmatrix} [\theta_j,\gamma_j] + \sigma^2\|\theta_j\| \\
&= \theta_j^\top\boldsymbol{\Sigma_{p_0}}\theta_j + 2\theta_j^\top C_{\boldsymbol{xU}}\gamma_j + \gamma_j^\top\boldsymbol{\Sigma_U}\gamma_j + \sigma^2\|\theta_j\|
\end{aligned}$$

Standardizing l_j we get $l_j = \tilde{m_j} + \tilde{S_j}t$ where $t \sim \mathcal{N}(0,1)$. Now by Stein's lemma applying to $(\boldsymbol{x_0}, l_j)$ jointly Gaussian:

$$\begin{aligned}
\text{Cov}(\boldsymbol{x_0},\phi^{\boldsymbol{U}})_{:j} &= \mathbb{E}_{\boldsymbol{x_0},\boldsymbol{Z},\boldsymbol{U}}[(\boldsymbol{x_0} - \mu_{\boldsymbol{p_0}})(\omega(l_j)] \\
&= \text{Cov}(\boldsymbol{x_0}, l_j)\mathbb{E}[\omega'(l_j)] \\
&= (\boldsymbol{\Sigma_{p_0}}\theta_j + C_{\boldsymbol{xU}}\gamma_j)\mathbb{E}[\omega'(l_j)]
\end{aligned}$$

Rewrite $\omega(l_j) = \omega(\tilde{m_j} + \tilde{S_j}t) =: g(t)$, so by Stein's lemma again:

$$\mathbb{E}_{t\sim\mathcal{N}(0,1)}[g'(t)] = \mathbb{E}_{t\sim\mathcal{N}(0,1)}[t \cdot g(t)]$$

So:

$$\begin{aligned}
\mathbb{E}_{t\sim\mathcal{N}(0,1)}[\tilde{S_j}\omega'(\tilde{m_j} + \tilde{S_j}t)] &= \mathbb{E}_{t\sim\mathcal{N}(0,1)}[\text{He}_1 \cdot \omega(\tilde{m_j} + \tilde{S_j}t)] \\
\tilde{S_j}\mathbb{E}_{l_j\sim\mathcal{N}(\tilde{m_j},\tilde{S_j^2})}[\omega'(l_j)] &= c_1(\tilde{m_j}, \tilde{S_j})
\end{aligned}$$

Therefore:

$$\mathbb{E}_{l_j\sim\mathcal{N}(\tilde{m_j},\tilde{S_j^2})}[\omega'(l_j)] = \frac{c_1(\tilde{m_j}, \tilde{S_j})}{\tilde{S_j}}$$

and:

$$\text{Cov}(\boldsymbol{x_0},\phi^{\boldsymbol{U}})_{:j} = (\boldsymbol{\Sigma_{p_0}}\theta_j + C_{\boldsymbol{xU}}\gamma_j)\alpha_j \qquad \text{where } \alpha_j = \frac{c_1(\tilde{m_j}, \tilde{S_j})}{\tilde{S_j}}$$

Meanwhile, we consider $\boldsymbol{\Sigma}_{\phi,ij}^{\boldsymbol{U}}$; first standardizing each of l_i and l_j is:

$$t_i = \frac{l_i - \tilde{m_i}}{\tilde{S_i}} \sim \mathcal{N}(0,1) \qquad t_j = \frac{l_j - \tilde{m_j}}{\tilde{S_j}} \sim \mathcal{N}(0,1)$$

and their correlation is:

$$\tilde{r}_{ij} = \text{Corr}(t_i, t_j) = \frac{\theta_i^\top\boldsymbol{\Sigma_{p_0}}\theta_j + \theta_i^\top C_{\boldsymbol{xU}}\gamma_j + \gamma_i^\top C_{\boldsymbol{Ux}}\theta_j + \gamma_i^\top\boldsymbol{\Sigma_U}\gamma_j + \sigma^2\theta_i^\top\theta_j}{\tilde{S_i}\tilde{S_j}}$$

Hermite-expansion:

$$\omega(l_i) = \omega(\tilde{m}_i + \tilde{S}_i t_i) = \sum_{n=0} c_n(\tilde{m}_i, \tilde{S}_i)\mathrm{He}_n(t_i)$$
$$\omega(l_j) = \omega(\tilde{m}_j + \tilde{S}_j t_j) = \sum_{m=0} c_m(\tilde{m}_j, \tilde{S}_j)\mathrm{He}_m(t_j)$$

Then:

$$\begin{aligned}\mathbb{E}[\phi_i^{\boldsymbol{U}}\phi_j^{\boldsymbol{U}}] &= \sum_{n,m} c_n(\tilde{m}_i, \tilde{S}_i)\mathrm{He}_n(t_i)c_m(\tilde{m}_j, \tilde{S}_j)\mathrm{He}_m(t_j)\mathbb{E}\big[\mathrm{He}_n(t_i)\mathrm{He}_m(t_j)\big] \\ &= \sum_n n!\tilde{r}_{ij}^n c_n(\tilde{m}_i, \tilde{S}_i)c_n(\tilde{m}_j, \tilde{S}_j)\end{aligned}$$

And so:

$$\begin{aligned}\boldsymbol{\Sigma}_{\phi,ij}^{\boldsymbol{U}} &= \mathbb{E}[\phi_i^{\boldsymbol{U}}\phi_j^{\boldsymbol{U}}] - \mathbb{E}[\phi_i^{\boldsymbol{U}}]\mathbb{E}[\phi_j^{\boldsymbol{U}}] \\ &= \sum_n n!\tilde{r}_{ij}^n c_n(\tilde{m}_i, \tilde{S}_i)c_n(\tilde{m}_j, \tilde{S}_j) - c_0(\tilde{m}_i, \tilde{S}_i)c_0(\tilde{m}_j, \tilde{S}_j) \\ &= \sum_{n\geq 1} n!\tilde{r}_{ij}^n c_n(\tilde{m}_i, \tilde{S}_i)c_n(\tilde{m}_j, \tilde{S}_j)\end{aligned}$$

This only has 1 term instead of 2 like in the general case, since l_i and l_j are both Gaussians, so we can write a closed-form for their covariance instead of having to decompose it out like in (**??**).

So writing $\boldsymbol{A}_{\boldsymbol{U}} = (\boldsymbol{\Sigma}_{\boldsymbol{p_0}}\Theta^\top + C_{\boldsymbol{xU}}\Gamma^\top)\mathrm{diag}(\alpha)$, then:

$$\mathcal{L}_{\sigma,\boldsymbol{U}} = \mathrm{Tr}(\boldsymbol{\Sigma}_{\boldsymbol{p_0}}) - \mathrm{Tr}(\boldsymbol{A}_{\boldsymbol{U}}(\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}})^{-1}\boldsymbol{A}_{\boldsymbol{U}}^\top)$$

Similarly for the unconditional case:

$$\begin{aligned}m_j &= \theta_j\mu_{\boldsymbol{x_0}} \\ S_j^2 &= \theta_j^\top\boldsymbol{\Sigma}_{\boldsymbol{p_0}}\theta_j + \sigma^2\|\theta_j\|\end{aligned}$$

and:

$$\mathrm{Cov}(\boldsymbol{x_0}, \phi)_{:j} = (\boldsymbol{\Sigma}_{\boldsymbol{p_0}}\theta_j)\alpha_j^0 \qquad \text{where } \alpha_j^0 = \frac{c_1(m_j, S_j)}{S_j}$$
$$\boldsymbol{\Sigma}_{\phi,ij} = \sum_{n\geq 1} n!r_{ij}^n c_n(m_i, S_i)c_n(m_j, S_j)$$

And so:

$$\mathcal{L}_\sigma - \mathcal{L}_{\sigma,\boldsymbol{U}} = \mathrm{Tr}(\boldsymbol{A}_{\boldsymbol{U}}(\boldsymbol{\Sigma}_\phi^{\boldsymbol{U}})^{-1}\boldsymbol{A}_{\boldsymbol{U}}^\top) - \mathrm{Tr}(\boldsymbol{A}(\boldsymbol{\Sigma}_\phi)^{-1}\boldsymbol{A}^\top)$$

where $\boldsymbol{A} = \boldsymbol{\Sigma}_{\boldsymbol{p_0}}\Theta^\top\mathrm{diag}(\alpha^0)$